JEEVAN NEUPANE

AI/ML ENGINEER | SOFTWARE ENGINEER | CO-FOUNDER & CTO

+977-9745956296 | jeevan.neupane003@gmail.com | Kathmandu, Nepal

Portfolio | GitHub | LinkedIn

PROFILE

Computer Engineering graduate with hands-on experience in AI/ML, full-stack development, and medical AI research. Currently building healthcare AI systems at A.I Lab, with work spanning Nepali-English speech recognition, clinical note generation, backend integration, and model deployment.

EXPERIENCE

Co-founder & Chief Technology Officer — 2025 - Present

A.I Lab Pvt. Ltd. | Kathmandu, Nepal

- Working on Shruti, a voice-first clinical documentation system for doctor-patient conversations.
- Developing and evaluating Nepali-English medical speech recognition, diarization, and LLM pipelines for structured clinical notes.
- Working on backend APIs, model deployment, EMR integration, and pilot testing with healthcare partners and clinics.

Fullstack Intern (Hybrid) — Feb 2026 - Apr 2026

Fusemachines

- Developed a Learning Management System for the Fuse AI Fellowship across frontend, backend, and database components.
- Built system features and APIs and performed testing and quality checks for reliable integration.

Research Intern (Part Time) — Feb 2025 - Jan 2026

NAAMII

- Worked on X-ray to 3D bone reconstruction using DRR generation, mesh deformation, CNNs, and transformer-based models.
- Developed retinal disease classification models using EfficientNet, ResNet, and ensemble methods for multi-label and multi-class prediction.

Software Engineering Intern (Part Time) — Dec 2024 - Dec 2025

Institute of Engineering (IOE)

- Assisted with the frontend of an examination and result management system during the transition to a new syllabus.
- Prepared technical documentation for system roles and workflows.

RESEARCH & PUBLICATION

Accepted Poster Paper | MIRASOL 2026 (MICCAI Workshop) — **2026**

"Large-Scale Pretraining for Improving Deep Learning-Based Geometric Distortion Correction of Diffusion-Weighted Imaging."

EDUCATION

Bachelor of Computer Engineering — Apr 2022 - Apr 2026

Pulchowk Campus, Tribhuvan University

Relevant coursework: Artificial Intelligence, Data Structures & Algorithms, Database Systems, Operating Systems, Software Engineering, Computer Networks, Embedded Systems, Computer Organization & Architecture.

SELECTED PROGRAMS & ACHIEVEMENTS

Selected Participant | IN-SPAN 2nd Cohort - Embassy of India & IIT Madras Pravartak Foundation **Jun - Jul 2026**

Completed an 8-week residential program in Chennai with AI-for-leaders sessions, industry CXO interactions, and visits to the IIT Madras Research Park ecosystem, including AI4Bharat and CHAI.

Selected Participant | APIE Camp #06, SOI Asia & Universitas Brawijaya (Indonesia) **Feb 2025**

Selected as a top-performing student for the international APIE Camp program.

Winner | Data Science and AI Track, KU Hackfest 2024 **Nov 2024**

1st place among 9 teams for an AI solution.

Winner | Best AI Integration, Hackademia 2.0 **Sep 2024**

1st place among 17 teams for AI implementation.

Winner | Best Presentation, Hackademia 1.0 **Nov 2023**

Recognized for the most impactful presentation.

SELECTED PROJECTS

X-Ray to 3D Bone Reconstruction Link

Implemented TransVert CNN and SwinX2S transformer approaches and generated DRR data for reconstructing 3D bone shapes from X-ray images.

LOCUS Registration & Gaming System Link

Built a registration and gaming platform for LOCUS 2026 supporting about 2,000 participants, including registration, score tracking, and live leaderboards.

Human Action Recognition Link

Implemented and fine-tuned transformer-based video models on UCF-101 and NTU RGB+D 120 and created a custom 10-action dataset.

VITON - Virtual Try-On

Built a deep-learning-based virtual try-on prototype and an interactive web interface for digital outfit preview.

TECHNICAL SKILLS

Programming: Python, C/C++, JavaScript, TypeScript
AI/ML: PyTorch, Hugging Face, NVIDIA NeMo, Scikit-learn, NumPy, Pandas, Matplotlib
Speech & LLM: ASR, code-mixed speech, diarization, medical note generation, LLM pipelines
Frontend: React.js, Next.js, Redux
Backend: Node.js, Express.js, FastAPI, Nest.js, Socket.io
Databases: MongoDB, MySQL
Systems & Cloud: Linux (Ubuntu/Debian), AWS (basic), GPU model deployment

CERTIFICATIONS

Certificate in Artificial Intelligence - Samsung Innovation Campus **Dec 2025**

Data science fundamentals, AI mathematics, data preprocessing, NLP, and computer vision.

Microdegree in Artificial Intelligence - Fusemachines **Dec 2025**

Generative AI, machine learning, computer vision, NLP, deep learning, and MLOps.

Operating the Internet - Keio University & APIE **Nov 2024**

Network planning, IP addressing, subnetting, NAT, DNS, security, and troubleshooting.

Introduction to Open Source - IBM **Sep 2024**

Open source fundamentals, GitHub, licensing, collaboration, and governance.

COMMUNITY INVOLVEMENT

Team Lead | LOCUS Open Source Team **Jul 2025 - Present**

Led a team developing software projects for the annual LOCUS tech festival.

Secretary | AWS Cloud Club IOE Pulchowk **May 2025 - Present**

Coordinated events, documentation, and team activities.